

Fixfse cluing a:

MACHINE LEARNING PROJECT

MNIST Handwritten Digit Classification

Using a Bidirectional Long Short-Term Memory neural network

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PROJECT SCOPE

The project turns handwritten pixels into digit predictions

Recognize handwritten digits from 0 to 9 using a recurrent neural network.

INPUT

Each 28 × 28 grayscale image becomes 28 sequential rows of 28 normalized pixel values.

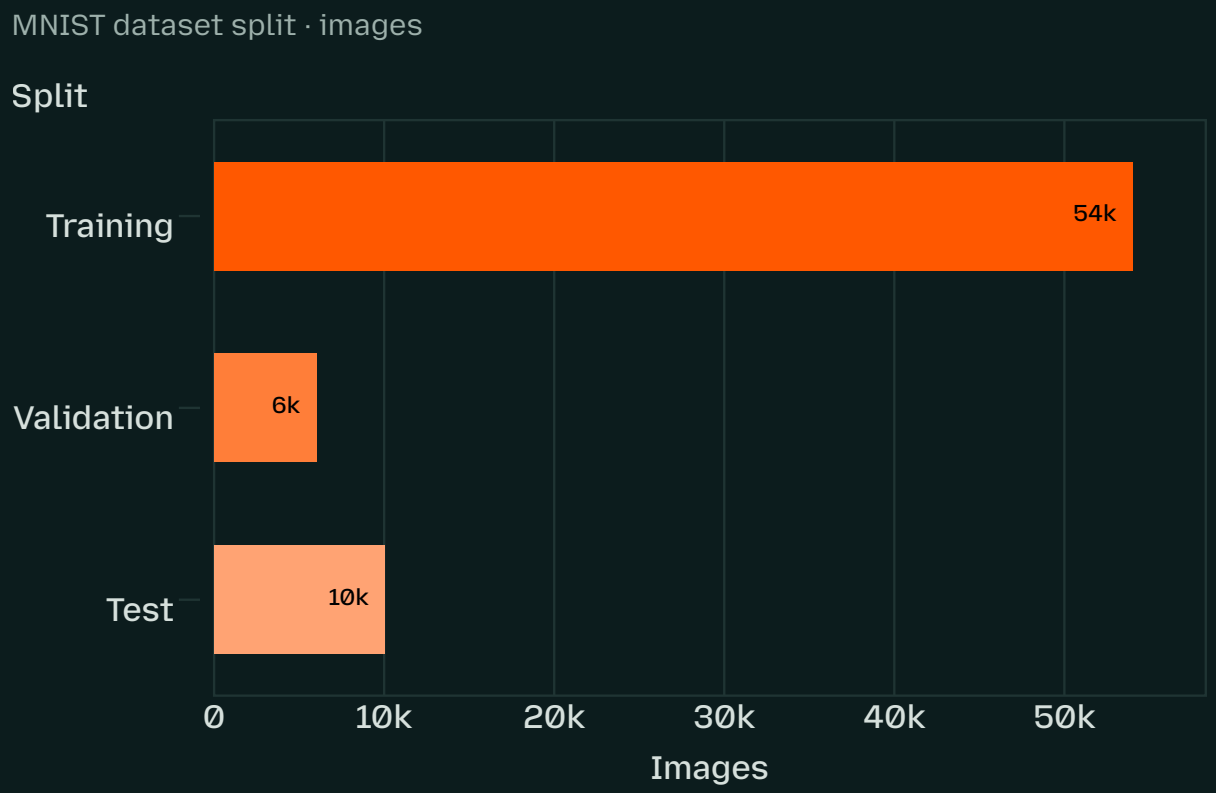
WORKFLOW

Run the complete workflow in Google Colab with TensorFlow/Keras.

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MNIST provides 70,000 labeled examples for a ten-class problem

The dataset is dominated by training examples, with a separate validation slice and a held-out test set.



10% of the training portion is used for validation during fitting.

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Row-wise sequencing makes an image readable to an LSTM

A 28 × 28 image becomes a sequence without losing its pixel structure.

01

Normalize

Normalize grayscale pixel values to the 0–1 range.

02

Sequence

Represent each image as 28 time steps.

03

Feed

Feed 28 pixel features at every step.

04

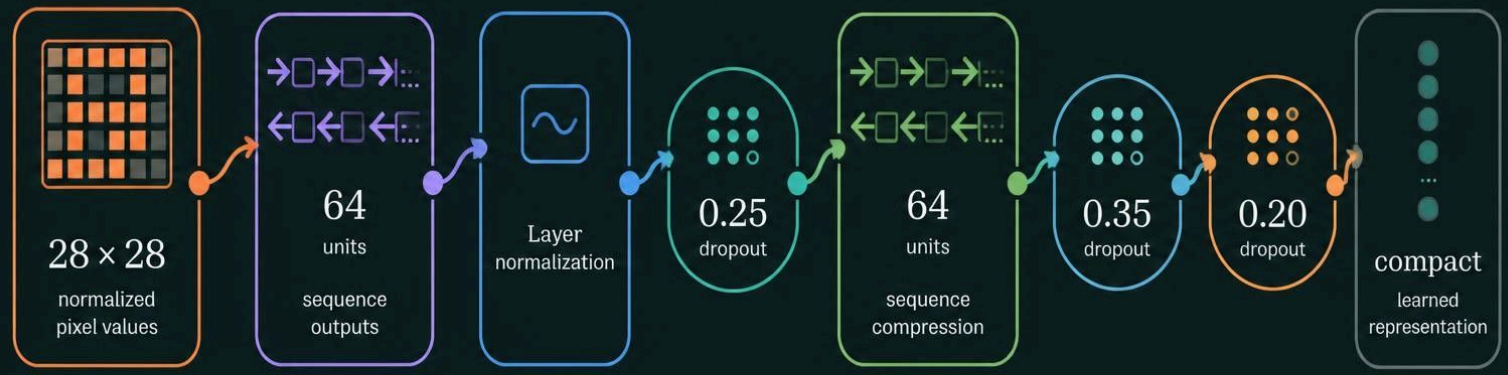
Learn

Learn patterns across the full image sequence.

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MODEL CORE

Bidirectional memory captures context from both directions



Sequence outputs move through normalization, dropout, and a compact representation.

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The classifier converts learned features into ten probabilities

The final layers turn the compact representation into a prediction from 0 to 9.

01 FEATURES

Compact representation

Learned features arrive from the BiLSTM pipeline.

02 DENSE

64 units

ReLU activation transforms the learned representation.

03 SOFTMAX

10 outputs

The highest probability becomes the predicted digit from 0 to 9.

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Training balances learning speed with generalization

The training setup combines an adaptive optimizer, controlled batches, and a validation-based learning-rate adjustment.

01

Optimize

Adam optimizer with sparse categorical cross-entropy.

02

Fit

10 epochs and batch size 128.

03

Track

Accuracy is the tracked metric.

04

Adapt

ReduceLROnPlateau lowers the learning rate when validation loss stops improving.

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Evaluation checks accuracy, errors, and confidence

Each view answers a different question about how the classifier behaves.



Learning curves

Track training and validation accuracy/loss curves.



Class report

Check precision, recall, and F1-score by digit.



Confusion matrix

Reveal commonly confused digits.

Examples

Compare actual-versus-predicted examples visually.

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Results should be filled from the executed notebook

The documentation describes the evaluation outputs but does not include the notebook’s numeric run results.

TEST ACCURACY

[Insert measured value]

TEST LOSS

[Insert measured value]

KEY CONFUSION

[Insert the most frequent digit pair from the matrix]

NEXT COMPARISON

Evaluate a CNN baseline on the same test set.

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The project produces a reusable model and prediction file

The final artifacts preserve both the trained classifier and the row-level predictions needed for review.

PREDICTIONS

mnist_lstm_predictions.csv

Actual digit, predicted digit, correctness flag, and confidence.

MODEL

mnist_bidirectional_lstm.keras

Saved trained Keras model.

REPRODUCIBILITY

Google Colab

Workflow is reproducible by running cells in order.

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